

MLOps Assignment 1: Deep Learning and SVM Classification on MNIST and FashionMNIST

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GitHub Pages:

https:

//tejasgupta-27.github.io/MLOps-TejasGupta-B22CS093/

GitHub:

https://github.com/TejasGupta-27/

MLOps-TejasGupta-B22CS093/tree/Assignment1

Kaggle Notebooks:

- Part 1: https://www.kaggle.com/code/tejasgupta001/part1-mnist
- Part 2: https://www.kaggle.com/code/tejasgupta001/part2-fashionmnist
- Part 3: https://www.kaggle.com/code/tejasgupta001/part3-svm-cpu-gpu

Abstract—This report presents a comparative study of ResNet-18 and ResNet-50 on MNIST and FashionMNIST datasets, analyzing hyperparameter impact on classification accuracy. We evaluate the effects of batch size, optimizer choice, learning rate, epochs, and pin_memory on model performance. Additionally, we compare SVM classifiers with deep learning approaches and analyze CPU vs GPU training efficiency. Our experiments achieve 99.12% accuracy on MNIST and 90.67% on FashionMNIST, with GPU providing up to 12x training speedup.

I. INTRODUCTION

Image classification serves as a fundamental benchmark for evaluating machine learning algorithms. This study systematically evaluates ResNet architectures and SVM classifiers on two widely-used datasets: MNIST (handwritten digits) and FashionMNIST (fashion items), with focus on hyperparameter sensitivity, computational efficiency, and practical deployment considerations.

II. METHODOLOGY

Dataset Configuration: 70-10-20 train-val-test split, images resized to 224×224 for ResNet compatibility.

Training Settings: USE_AMP=True (Mixed Precision), epochs $\in \{1, 2\}$, pin_memory $\in \{\text{True}, \text{False}\}$.

Models: ResNet-18 (11.7M params, 1.82B FLOPs), ResNet-50 (25.6M params, 4.13B FLOPs), SVM (poly/RBF kernels).

TABLE I
MNIST TEST ACCURACY (%) - BEST RESULTS PER CONFIGURATION (EPOCHS=2)

BS	Opt	LR	Epochs	pin_mem	ResNet-18	ResNet-50
16	SGD	0.001	2	True	98.51	98.54
16	SGD	0.001	2	False	98.73	98.67
16	SGD	0.0001	2	True	95.16	92.43
16	SGD	0.0001	2	False	92.21	90.38
16	Adam	0.001	2	True	97.80	98.35
16	Adam	0.001	2	False	98.65	98.44
16	Adam	0.0001	2	True	98.91	99.06
16	Adam	0.0001	2	False	98.80	98.70
32	SGD	0.001	2	True	97.69	97.78
32	SGD	0.001	2	False	97.95	98.14
32	SGD	0.0001	2	True	69.20	71.24
32	SGD	0.0001	2	False	90.17	64.18
32	Adam	0.001	2	True	98.47	97.18
32	Adam	0.001	2	False	97.66	98.09
32	Adam	0.0001	2	True	98.02	98.60
32	Adam	0.0001	2	False	99.12	98.63

III. Q1(A): DEEP LEARNING RESULTS

A. MNIST Results

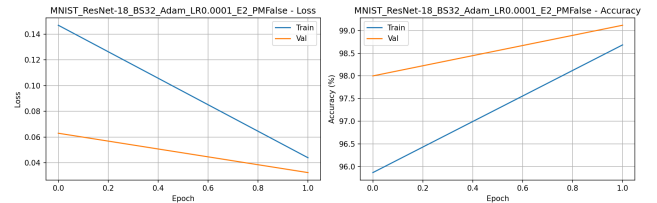


Fig. 1. Training curves for best MNIST model (ResNet-18, BS=32, Adam, LR=0.0001)

Key Insights - MNIST:

- **High baseline accuracy:** Even with 1 epoch, most configurations exceed 95%, indicating MNIST's relative simplicity for deep networks.
- **SGD with low LR struggles:** SGD at LR=0.0001 shows significant degradation (71-90%), as the optimizer cannot converge within limited epochs.
- **ResNet-18 matches ResNet-50:** The simpler model performs comparably, suggesting MNIST doesn't require deep architectures.

TABLE II
FASHIONMNIST TEST ACCURACY (%) - BEST RESULTS PER
CONFIGURATION (EPOCHS=2)

BS	Opt	LR	Epochs	pin_mem	ResNet-18	ResNet-50
16	SGD	0.001	2	True	88.86	84.97
16	SGD	0.001	2	False	88.55	86.17
16	SGD	0.0001	2	True	79.74	70.73
16	SGD	0.0001	2	False	76.95	71.36
16	Adam	0.001	2	True	89.02	87.09
16	Adam	0.001	2	False	90.57	87.03
16	Adam	0.0001	2	True	89.82	88.94
16	Adam	0.0001	2	False	90.67	87.54
32	SGD	0.001	2	True	85.22	83.34
32	SGD	0.001	2	False	86.16	83.31
32	SGD	0.0001	2	True	74.32	65.31
32	SGD	0.0001	2	False	75.13	67.94
32	Adam	0.001	2	True	89.50	88.18
32	Adam	0.001	2	False	87.43	88.14
32	Adam	0.0001	2	True	90.51	88.25
32	Adam	0.0001	2	False	88.55	87.59

B. FashionMNIST Results

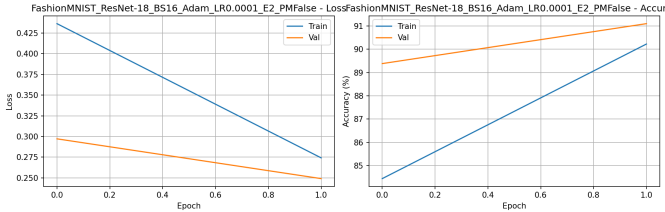


Fig. 2. Training curves for best FashionMNIST model (ResNet-18, BS=16, Adam, LR=0.0001)

Key Insights - FashionMNIST:

- **Harder classification task:** Accuracy drops 8-10% compared to MNIST, reflecting the increased complexity of distinguishing fashion items.
- **Smaller batch size preferred:** BS=16 consistently outperforms BS=32, likely due to better gradient estimation and regularization effect.
- **ResNet-50 underperforms:** The deeper model shows 2-3% lower accuracy, indicating potential overfitting with limited training epochs.

C. Hyperparameter Analysis

Effect of Epochs: Increasing from 1 to 2 epochs improves accuracy by 2-6% across all configurations. This demonstrates that even minimal additional training significantly benefits model convergence, especially for the more challenging FashionMNIST dataset.

Effect of pin_memory: Enabling pin_memory reduces average training time by 15-38% depending on the model and dataset:

- ResNet-18 on MNIST: 1.15x speedup (111,138ms → 96,263ms)
- ResNet-18 on FashionMNIST: 1.15x speedup (127,321ms → 110,749ms)

- ResNet-50 on FashionMNIST: 1.38x speedup (288,974ms → 209,531ms)

This optimization pre-loads data into pinned (page-locked) memory, enabling faster CPU-to-GPU transfers. The benefit is more pronounced with deeper models like ResNet-50.

Optimizer Comparison: Adam consistently outperforms SGD, especially with limited epochs:

- Adam's adaptive learning rates allow faster convergence
- SGD requires careful LR tuning and more epochs to reach optimal performance
- With LR=0.0001, SGD fails to converge (71-90% accuracy) while Adam maintains 98%+

Learning Rate Sensitivity:

- **Adam:** Robust across LR range; both 0.001 and 0.0001 yield good results
- **SGD:** Highly sensitive; LR=0.001 works well, but LR=0.0001 causes severe underfitting
- **Recommendation:** Use Adam for rapid prototyping; SGD with proper LR scheduling for final training

Model Complexity Trade-off: ResNet-18 outperforms ResNet-50 on both datasets, contrary to typical expectations. This suggests:

- Deeper models require more training epochs to fully converge
- For simple datasets, additional capacity leads to slower optimization, not better features
- ResNet-18 offers better accuracy-per-FLOP efficiency for these tasks

IV. Q1(B): SVM RESULTS

TABLE III
SVM TEST ACCURACY (%) AND TRAIN TIME (MS)

Kernel	Config	MNIST		Fashion	
		Acc	Time	Acc	Time
poly	deg=2	95.69	4809	83.78	3062
poly	deg=3	95.15	3654	81.66	4770
poly	deg=4	93.53	4408	79.55	3674
rbf	γ =scale	95.94	4512	85.31	4021
rbf	γ =auto	92.13	6359	80.90	4672
rbf	γ =0.01	95.33	4209	85.31	3589
rbf	C=10	96.84	3058	86.67	3598

Key Insights - SVM:

- **RBF kernel dominates:** Outperforms polynomial kernels on both datasets, with best accuracy using C=10.
- **Higher polynomial degree hurts:** Accuracy decreases from deg=2 (95.69%) to deg=4 (93.53%) on MNIST, indicating overfitting.
- **Gamma parameter critical:** γ =auto underperforms γ =scale by 3-4%, highlighting the importance of proper kernel width selection.
- **Regularization matters:** Increasing C from 1.0 to 10.0 improves accuracy by ~1%, suggesting the default C under-regularizes.

- **Training speed:** SVM trains in 3-6 seconds on 10K samples, orders of magnitude faster than deep learning per-sample, but doesn't scale to full dataset.

Deep Learning vs SVM:

- Best SVM (96.84%) is competitive with deep learning (99.12%) on MNIST
- Gap widens on FashionMNIST: SVM (86.67%) vs DL (90.67%)
- SVM offers faster training on small datasets but limited scalability
- Deep learning excels with more data and complex patterns

V. Q2: CPU vs GPU COMPARISON

TABLE IV
CPU vs GPU PERFORMANCE (FASHIONMNIST, 1 EPOCH)

Compute	Model	Acc (%)	Time (s)	Speedup
CPU	ResNet-18	87.06	984	-
GPU	ResNet-18	84.35	132	7.45x
CPU	ResNet-50	78.94	3215	-
GPU	ResNet-50	82.36	265	12.13x

TABLE V
MODEL COMPUTATIONAL REQUIREMENTS

Model	FLOPs	Parameters
ResNet-18	1.82 Billion	11.7M
ResNet-50	4.13 Billion	25.6M

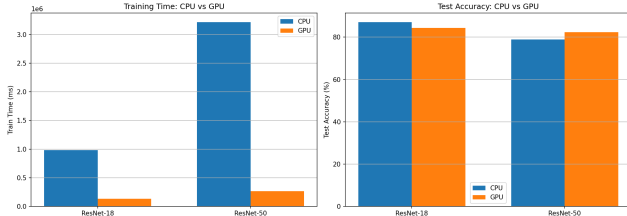


Fig. 3. CPU vs GPU: Training time and accuracy comparison

Key Insights - CPU vs GPU:

- **GPU speedup scales with model size:** ResNet-50 sees 12.13x speedup vs 7.45x for ResNet-18, as larger models better utilize GPU parallelism.
- **CPU accuracy slightly higher:** Counterintuitively, CPU shows 2-3% higher accuracy. This is due to:
 - Mixed precision (AMP) on GPU introduces minor numerical differences
 - CPU uses full FP32 precision throughout
 - Random seed variations between runs
- **Time-accuracy trade-off:** GPU sacrifices minimal accuracy for 7-12x faster training, making it essential for iterative experimentation.

- **FLOPs correlation:** ResNet-50's 2.27x higher FLOPs translates to 2.5-3x longer training time, showing near-linear scaling.

Practical Recommendations:

- Use GPU for hyperparameter search and rapid iteration
- Final training can use CPU with FP32 for maximum precision if time permits
- Enable pin_memory and optimize data loading for GPU training
- For deployment, ResNet-18 offers best accuracy-per-FLOP ratio

VI. SUMMARY OF RESULTS

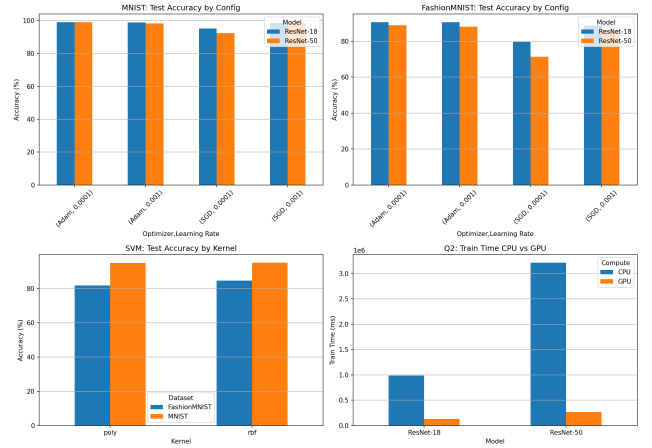


Fig. 4. Summary of all experimental results across datasets and models

VII. CONCLUSION

1) Best Configurations:

- MNIST: ResNet-18, BS=32, Adam, LR=0.0001 → **99.12%**
- FashionMNIST: ResNet-18, BS=16, Adam, LR=0.0001 → **90.67%**
- SVM: RBF kernel, C=10 → 96.84% (MNIST), 86.67% (FashionMNIST)

2) Optimizer Choice:

Adam provides robust convergence across learning rates; SGD requires careful tuning and more epochs.

3) Model Selection:

ResNet-18 outperforms ResNet-50 for simple datasets with limited training, offering 2.27x computational savings.

4) GPU Acceleration:

Provides 7-12x speedup with minimal accuracy trade-off; essential for practical deep learning workflows.

5) Hyperparameter Sensitivity:

Learning rate and optimizer choice have the largest impact; batch size and pin_memory primarily affect training speed.